

Exam 2019/20

Question 1

Q: What was the first (recorded) software bug and how was it identified? Please justify your answer.

- a) On the 50s, using NATO's advanced bug identification systems
- b) On the 80s, using critical service analysis
- c) On the 40s, manually
- d) On the 90s, using automated testing.

A: c).

Back in the 40s, computers filled large rooms and their warmth attracted bugs. These creatures then shortened circuits and caused the computer to malfunction, which lead, on September 9, 1945, U.S. Navy officer Grace Hopper to manually find a moth between the relays on the Harvard Mark II computer she was working on.

Question 2

Q: Which one of the following is most adequate to have a software prototype (please justify your answer):

- a) A customer-driven app for ordering food;
- b) A business-to-business software for supporting a supply chain;
- c) A healthcare software to handle critical data;
- d) All of the above

A: d)

All of the above options should include prototyping activities, as this is a crucial step to help elicit requirements for later increments. Early increments of the process should be used to better study the problem in question, by establishing a Prototyping plan, outlining its definition, executing and then evaluating on the result.

Question 3

Q: With RUP, the construction phase is when the actual software is most actively developed. Identify the disciplines that are most active during this phase and briefly explain why.

- a) All disciplines are equally active;
- b) The only active discipline is the implementation;
- c) Implementation and testing;
- d) Design, implementation and testing.

A: d)

As mentioned, the construction phase is when the software is most actively developed, the Implementation. Along with the development, it is thoroughly tested against the defined evaluation criteria and the product releases are assessed against acceptance criteria for the vision, reason why Testing also represents a big part in this phase. Additionally, there should exist a concern with resource management, control and process optimization. This analysis is important and may lead to design changes, making the Analysis & Design an active discipline.

Question 4

Q: XP actively prevent software regressions by motivating a test first approach. Who should participate in this process? Justify your choice.

- a) The client should provide the development team with tests for his requirements;
- b) The client and the project manager should write tests before tackling new features;
- c) The development team, with input from the client, should write tests before tackling new features;
- d) The development team, with input from the project manager, should write tests before tackling new features;

A: c)

It is the development team that writes the tests before tackling new features. The customer should participate in the process by specifying acceptance tests, also called customer tests. A user story will be complete when all of its acceptance tests pass.

Unit Tests are used to test specific methods, classes or scenarios of the code and are internal to the development team.

Question 5

Q: Missing project deadlines is possibly the most recurring failure in software development. How does Scrum try to address it? Justify your answer.

- a) Scrum will only ensure that a product increment exists in the end of each sprint, but not the quantity of features that are part of it. Traditional methodologies are more strict hence, more capable of meeting deadlines;
- b) The Scrum Master is responsible for ensuring that the team is efficient and delivers the software at the right time;
- c) Effort estimations and velocity can be used to forecast when the backlog will be complete;
- d) A well defined process will ensure development efficiency, which will guarantee the project deadline.

A: c).

In each sprint the team's velocity and effort is measured in order to allow that, in the following sprints, that information can be used to forecast how much progress can be done. With this method, the team can have a better understanding of how they progress and prioritize the next tasks and requirements in a more informed and adequate way. While not completely certain, this method will ensure a higher change of meeting the necessary deadlines, allowing to guess when the backlog will be complete.

Question 6

Q: Brainstorming is a common requirement elicitation technique in which (justify your answer):

- a) the ideas only come from the system end users
- b) all the ideas are welcome
- c) there is no facilitator
- d) 15 people is the limit of people in the room.

A: b).

The idea with brainstorming is to elicit new and innovative requirements. There should be a moderator and around 4-8 people with different/multiple perspectives on the product, not just system end users. Additionally, all the ideas are welcome, no matter how creative or visionary they may sound, prioritizing quantity over quality. In a later step they are discussed, revised and organized.

Question 7

Q: Regarding use case modelling, include relationships aim to (justify your answer):

- a) Represent optional behavior
- b) Represent critical behavior
- c) Represent mandatory behavior
- d) Represent behavior imported from other system

A: c).

When several use cases share some common behaviour, that behaviour can be separated and described in a new use case which is included by the first ones. This common behaviour may or may not be imported from other system, but it is always mandatory and can not be removed.

Question 8

Q: Software Engineering has been a research subject for the past decades. Focusing on software architectures, engineers have learned much about how to build good software. What technique/approach is typically adopted to share such knowledge between engineers? Describe why is it a good technique/approach.

- a) Object Oriented Programming.
- b) Reusable Open Source libraries.
- c) Design Patterns.
- d) Sequence Diagrams.

A: c).

Design Patterns specify template solutions for common types of problems, describing how modules can/should be organized in order to maintain a good codebase. These are solutions that have worked well before and adapt smoothly and incrementally to new situations. They should be used instead of architecting solutions from scratch, as this is hard and, most times, not effective.

Question 9

Q: The Project Management Body of Knowledge states that the project management processes are divided into process groups. How many process groups are there and what are their names and main objective:

- a) 6
- b) 12
- c) 5
- d) 3

A: There are 5 process groups:

1. Initiating Process Group: active at the start of the project
2. Planning Process Group: responsible for the project planning, always active as it is important
3. Executing Process Group: active in the development phases

Question 10

Q: Reverse engineering (please justify your answer):

- a) Is a reengineering process activity
- b) Means converting code to a new language
- c) Is to clean-up and restructure system data
- d) Is to reorganise the program structure.

A: a).

Reverse engineering is a reengineering process where we take an existing program or codebase and analyse it, in order to better understand it. This can be done in multiple ways, either by running it and see how it behaves or just statically read the code. Just like the other reengineering processes, this method will help with re-structuring and re-documenting software to make it easier to understand and change.